

Role-Based Authentication (RBAC)

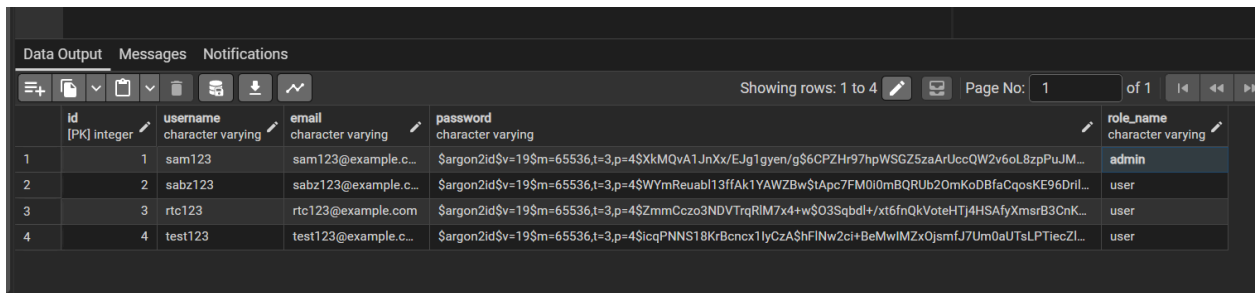
Today, I implemented **Role-Based Access Control (RBAC)** in the project. The system now supports two roles:

- **Admin**
- **User**

The access to routes and functionalities is controlled based on these roles.

Role-Based Authentication Setup

- Created a **Role** table
- Manually inserted roles (**admin**, **user**) into the database
- After first registration, I manually changed the first user's role to **admin**



	Id [PK] integer	username character varying	email character varying	password character varying	role_name character varying
1	1	sam123	sam123@example.c...	\$argon2id\$v=19\$m=65536,t=3,p=4\$XkMQvA1JnXx/EJg1gyen/g\$6CPZHr97hpWSGZ5zaArUccQW2v6oL8zpPuJM...	admin
2	2	sabz123	sabz123@example.c...	\$argon2id\$v=19\$m=65536,t=3,p=4\$WYmReuabl13fAk1YAWZBw\$TApc7FM0i0mBQRUb2OmKoDBfaCqosKE96Dril...	user
3	3	rtc123	rtc123@example.com	\$argon2id\$v=19\$m=65536,t=3,p=4\$ZmmCczo3NDVTrqRIM7x4+w\$O3Sqbdl+/xt6fnQkVoteHTJ4HSAfyXmsrB3CnK...	user
4	4	test123	test123@example.c...	\$argon2id\$v=19\$m=65536,t=3,p=4\$icqPNNs18KrBcncx1lyCzA\$hFINw2ci+BeMwIMZxOjsmfJ7Um0aUTsLPTiecZL...	user

Admin access control

Resource:User

- **User Registration** – Accessible by both Admin(to create user) and User
- **Login System**— Accessible by both Admin and User
- **Show All Users** – Only Admin can view users
- **Delete User** – Only Admin can delete users

Resource:Project

- **Create Project** – Only Admin
- **Show All Projects** – Accessible by both Admin and User(see assigned project)
- **Show Single Project** – Accessible by both Admin and User(see assigned project)

- **Assign Project to User** – Admin assigns projects to users

Resource:tasks

- **Create Task** – Only Admin
- **Show Tasks-both**
- **Update Task Status** – Both Admin and User can update task (e.g., Pending → Completed)
- **Delete Task** – admin can delete task